CHEM-SHIELD

An AI-Assisted Cyber-Physical Early-Warning and Threat Detection Framework for Simulated Defense-Industrial Operational Technology

REVISED ACADEMIC RESEARCH PROPOSAL

Defensive Cybersecurity • Artificial Intelligence • OT/ICS Security

Implementation Scope: Controlled Virtual Laboratory Simulation

Target Context: Ethiopian Defense-Industrial Support Environment (Simulated)

Project Type: Defensive Cybersecurity Academic Research

Core Paradigm: Multi-Layer Cyber-Physical Event Detection, Correlation, and Risk Assessment

Scope Reduction Note

This revised proposal intentionally reduces the original implementation scope to a feasible laboratory prototype that can be completed with limited time and resources. The core research idea is preserved while non-essential infrastructure, advanced response automation, and oversized tooling are removed.

Executive Summary

CHEM-SHIELD is a defensive cybersecurity research project that investigates whether combining network-security evidence with physical-process anomaly detection can provide earlier and more useful warnings of suspicious activity in an Operational Technology (OT) environment. The revised project is implemented entirely inside an isolated virtual laboratory and does not access, scan, modify, or test real military, industrial, SCADA, PLC, or defense infrastructure.

The reduced implementation consists of five tightly connected components:

(1) OT simulator containing a virtual PLC, machine/process model, and sensors;

(2) lightweight network-security monitoring for the simulated OT traffic;

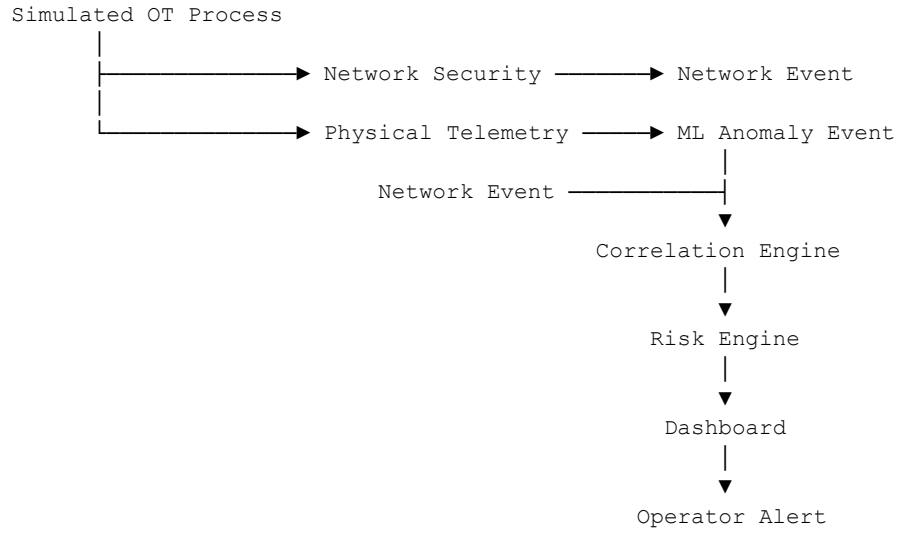
(3) a machine-learning anomaly detector that learns normal process behavior;

(4) a rule-based correlation and risk engine that combines network and physical evidence; and

(5) a Flask-based dashboard that presents machine state, security events, anomaly scores, correlated events, and risk levels.

The central research idea is intentionally simple: a simulated machine can be affected through the network, producing both network-level evidence and, potentially, abnormal physical behavior. Network monitoring detects suspicious communications, while machine learning detects deviations from the normal physical baseline. The correlation and risk layer determines whether the two evidence streams are related and prioritizes the resulting event. The dashboard then provides a clear early-warning notification to the operator.

Core Research Concept



1. Introduction and Background

Operational Technology systems connect digital communications with physical processes. Unlike conventional IT systems, OT environments operate machinery, sensors, controllers, and process equipment where a cyber event may affect the state of a physical process. This makes monitoring both the network and the process behavior important.

The proposed CHEM-SHIELD project studies this problem in a generalized and simulated defense-industrial support context. The project does not model or claim knowledge of proprietary architectures belonging to specific Ethiopian facilities. Instead, it uses a generic virtual OT environment with standard industrial communication concepts and synthetic process telemetry.

The research is motivated by a specific limitation of network-only monitoring: a command can be syntactically valid at the protocol level while producing an unusual physical state. A network detector may therefore see a legitimate protocol transaction, while a process-behavior model can identify the resulting deviation.

2. Ethiopian Defense-Industrial Context and Research Boundary

Ethiopia has publicly documented defense-industrial and engineering capabilities, and the country's broader technology ecosystem includes cybersecurity, artificial intelligence, automation, communications, and advanced manufacturing. These facts establish a relevant academic context for studying defensive OT cybersecurity.

However, publicly available information does not provide sufficient technical detail to justify assumptions about the internal network architectures, PLC vendors, SCADA versions, segmentation, security controls, or operational procedures of specific military facilities. CHEM-SHIELD therefore deliberately avoids representing any named facility as the actual target or reproducing a real operational architecture.

Research Boundary

- All experiments occur in an isolated virtual laboratory.
- Only synthetic or simulated machines, PLCs, sensors, network traffic, and telemetry are used.
- No operational military or commercial network is scanned, probed, accessed, or modified.
- No physical defense equipment, production equipment, or live support infrastructure is connected to the experiment.
- The project does not develop malware, weaponized exploit payloads, or offensive attack tools.
- Controlled abnormal scenarios are used only to evaluate defensive detection and correlation.

3. Problem Statement

Conventional network-security monitoring is valuable for detecting suspicious communication patterns, but it does not necessarily determine whether a valid OT command produces an abnormal physical process state. Conversely, a physical anomaly alone may have many explanations, including legitimate process variation or equipment behavior. The research problem is therefore to determine whether combining network evidence with physical-process anomaly evidence can improve the identification and prioritization of suspicious cyber-physical events.

```
Network-only view:
Network command → protocol appears valid → limited security context

Cyber-physical view:
Network command → process changes → physical anomaly → correlated evidence → prioritized
warning
```

4. Research Gap and Contribution

The project does not claim to invent intrusion detection, machine learning, SIEM concepts, anomaly detection, or risk scoring. These are established areas. The proposed contribution is a small, experimentally testable framework that combines these ideas in a single controlled OT simulation and evaluates the value of evidence fusion.

The revised research gap is therefore formulated as follows:

Can a lightweight cyber-physical detection pipeline that combines simulated OT network events with machine-learning process anomalies provide more informative and timely early warnings than either evidence source considered independently?

Expected Contribution

- A reproducible virtual OT environment for defensive experimentation.
- A normal-process baseline and lightweight ML anomaly detector.
- A network-event detection layer for the simulated OT network.
- A transparent rule-based correlation mechanism.
- A documented risk-scoring model for prioritizing correlated events.
- A unified dashboard demonstrating the complete detection pipeline.
- Experimental results comparing normal operation, network-only anomalies, physical-only anomalies, and correlated cyber-physical anomalies.

5. Research Objectives

5.1 General Objective

To design and experimentally evaluate a lightweight AI-assisted cyber-physical early-warning framework that detects suspicious network activity and abnormal OT process behavior, correlates the evidence, assigns a risk level, and presents the result through a unified security dashboard in an isolated virtual laboratory.

5.2 Specific Objectives

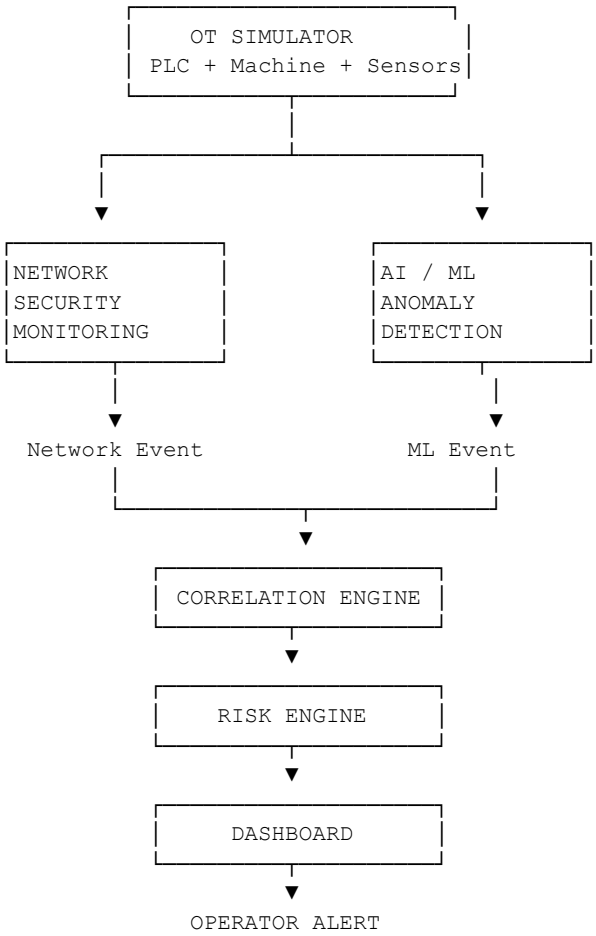
1. Develop a simulated OT environment consisting of a virtual PLC, process/machine model, sensors, and a simple HMI or control interface.
2. Generate and monitor simulated OT network traffic and identify suspicious network events.
3. Collect normal physical-process telemetry and use it to establish a baseline.
4. Implement a lightweight machine-learning anomaly detector, with Isolation Forest as the primary model.
5. Generate a normalized event format that allows network and ML components to exchange evidence.
6. Implement rule-based temporal and asset-aware event correlation.
7. Implement a transparent risk-scoring mechanism using network severity, physical anomaly evidence, asset criticality, and correlation strength.
8. Develop a single web dashboard for near-real-time monitoring and alert visualization.
9. Evaluate detection performance using precision, recall, F1-score, false-positive rate, and detection latency where measurable.

6. Research Questions

- RQ1: Can a simulated OT environment produce a reproducible normal behavioral baseline and controlled abnormal states?
- RQ2: Can network monitoring identify suspicious communication events in the simulated OT environment?
- RQ3: Can a lightweight machine-learning model distinguish normal process behavior from controlled anomalies?
- RQ4: Does correlating network and physical anomalies provide stronger evidence for prioritization than either source alone?
- RQ5: Can a transparent risk engine convert heterogeneous evidence into useful early-warning levels without requiring complex infrastructure?

7. Proposed CHEM-SHIELD Architecture

The revised architecture contains five implementation parts. An optional response-recommendation function may be added if time permits, but automated response is outside the core scope.



7.1 Part 1 OT Simulation

The OT simulator represents the physical environment in which the experiment occurs. It should contain a virtual PLC or controller, a simplified machine/process model, and a small set of virtual sensors. The

simulated process should produce variables such as RPM, temperature, pressure, target value, and machine status.

- PLC/controller: receives simulated control values and exposes process state.
- Machine/process model: converts control values into changing physical states.
- Sensors: report process variables.
- HMI/control interface: provides a simple way to observe or change normal operating parameters.
- Telemetry logger: records time-stamped physical states for ML training and evaluation.

The simulator does not need to reproduce a real defense-manufacturing process. A generic motor/process model is sufficient because the research question concerns detection and evidence correlation, not industrial process optimization.

7.2 Part 2 Network Security

The network-security layer observes communication inside the isolated OT simulation. Its purpose is to create structured network events that can later be correlated with physical anomalies.

- Monitor simulated OT traffic and communication between controller, HMI, and other virtual nodes.
- Identify suspicious or unauthorized control-related network events.
- Record event timestamp, source, destination, protocol, event type, and severity.
- Use a lightweight IDS/logging approach appropriate to the available resources.

Suricata and/or Zeek may be used where practical. The revised project does not require both to be fully deployed if resource or time constraints make one sufficient. The research requirement is the network-event evidence, not the number of security products.

7.3 Part 3 AI/ML Anomaly Detection

The ML component learns the normal behavior of the simulated physical process and identifies observations that deviate from that baseline. Isolation Forest is selected as the primary model because it is lightweight, available through Scikit-learn, and appropriate for a limited-resource anomaly-detection prototype.

Candidate input features include:

- RPM
- Temperature
- Pressure
- Target-versus-actual RPM deviation
- Rate of change of RPM
- Rate of change of temperature
- Rate of change of pressure

The ML system should produce two primary outputs: an anomaly status and an anomaly score. The model does not independently declare that an attack occurred. Instead, it reports that the observed process behavior is unusual. The correlation and risk layers then interpret that evidence in combination with network observations.

7.4 Part 4 Correlation and Risk Engine

The correlation engine combines network events and ML events. A simple rule-based approach is sufficient. Events may be considered related when they involve the same simulated asset, occur within a defined time window, and represent logically related behavior.

```
IF same asset
AND events occur within the correlation time window
AND network event indicates a relevant control/parameter change
AND physical event indicates abnormal process behavior
THEN create a correlated cyber-physical event.
```

The risk engine then prioritizes the correlated event. A transparent normalized score can incorporate network severity, physical anomaly severity, asset criticality, and correlation strength. The score is mapped to LOW, MEDIUM, or HIGH for the prototype.

The risk score is a research-design mechanism rather than a universal industrial risk standard. The selected weights and thresholds will be documented and kept consistent throughout the experiments.

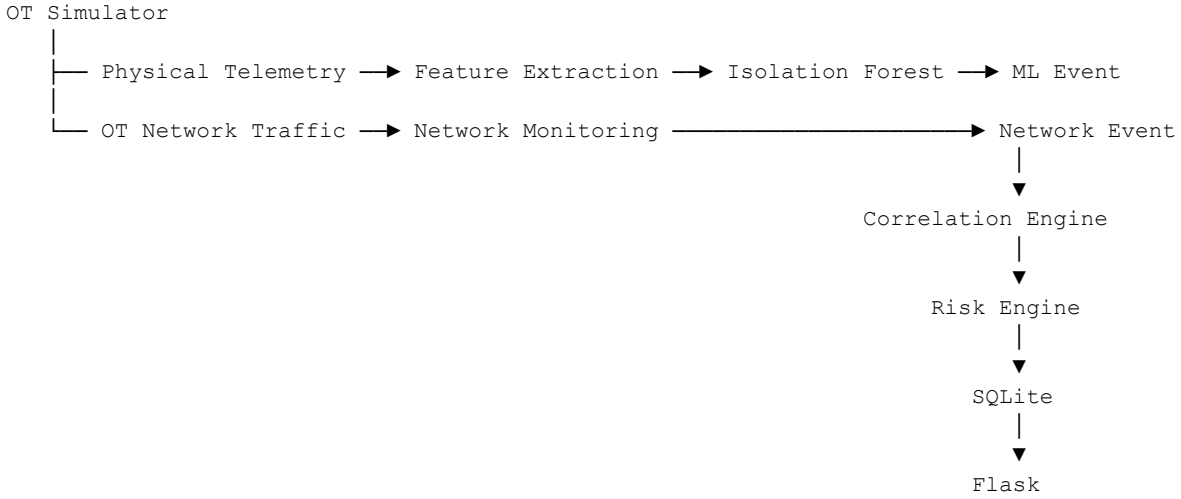
7.5 Part 5 Dashboard and Integration

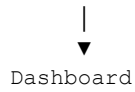
The dashboard provides the visible interface to the complete system. It should show current process state, network events, ML anomaly status, correlated events, risk score, and alert history.

- System status: PLC, machine, network monitor, ML engine, and risk engine.
- Current process state: RPM, temperature, pressure, target value, and machine status.
- Network event list.
- ML anomaly score and status.
- Correlated cyber-physical event details.
- Current risk level.
- Historical event table and basic time-series plots.
- Optional operator response recommendation for high-risk events.

Flask, HTML/CSS/JavaScript, and Plotly are sufficient. SQLite can be used as the initial event store. The dashboard is intentionally kept simple so that development effort remains focused on the research contribution.

8. End-to-End Data and Integration Flow





A unified event structure will be used so that the dashboard does not need to understand the internal implementation of each component.

```
{
  "timestamp": "...",
  "asset": "PLC-01",
  "event_type": "CYBER_PHYSICAL_ANOMALY",
  "network_anomaly": true,
  "physical_anomaly": true,
  "ml_score": 0.91,
  "correlation": "strong",
  "risk_score": 87,
  "risk_level": "HIGH"
}
```

9. Controlled Experimental Scenarios

The experiments are designed to demonstrate the difference between independent evidence sources and their correlation. The scenarios are performed only against the simulated environment.

Scenario	Network State	Physical State	Expected Interpretation
S1 Normal	Normal	Normal	Low risk; baseline behavior
S2 Network-only anomaly	Anomalous	Normal	Suspicious network event; investigate
S3 Physical-only anomaly	Normal	Anomalous	Process anomaly without corresponding network evidence
S4 Correlated cyber-physical event	Anomalous	Anomalous	Strong combined evidence; high-priority alert

10. Virtual Machine (OT Simulator)

The virtual machine, or OT simulator, is the foundation of CHEM-SHIELD. It represents a small industrial-style environment entirely in software so that the complete experiment can be performed safely without connecting to a real PLC, machine, SCADA system, or production network.

The simulator contains three main elements: a virtual PLC/controller, a simplified machine or process model, and virtual sensors. The PLC receives simulated control values, the machine/process model changes its state according to those values, and the sensors report variables such as RPM, temperature, pressure, target value, and machine status.

The purpose of the OT simulator is not to reproduce a real defense-industrial process exactly. Its purpose is to create a repeatable physical process in which normal behavior and controlled abnormal behavior can be generated. This gives the network-security and machine-learning components a common environment and makes the experiment reproducible.

- Virtual PLC/controller: manages simulated control values and exposes the current process state.
- Machine/process model: converts control inputs into changing physical values.

- Virtual sensors: generate time-stamped telemetry such as RPM, temperature, and pressure.
- HMI/control interface: provides a simple way to observe the process and change permitted simulation parameters.
- Telemetry logger: records the process data used for ML training, anomaly detection, correlation, and evaluation.

11. Network Security and OT Security

The network-security and OT-security part monitors communication inside the isolated simulated OT environment. Its purpose is to identify suspicious network behavior and convert that behavior into structured security evidence that can be compared with physical-process behavior.

In the CHEM-SHIELD model, the network layer observes communication between the virtual PLC, HMI, and other simulated nodes. It can record events such as unusual communication, unexpected control-related traffic, or suspicious parameter-change activity. Each event should include basic information such as timestamp, source, destination, protocol, event type, and severity.

Suricata and/or Zeek may be used where practical. If resources are limited, one lightweight monitoring component is sufficient. The important research requirement is that the network layer produces reliable, structured evidence for the correlation engine rather than simply adding more security tools.

This layer also represents the OT-security perspective: security events must be interpreted in relation to the industrial process and the asset being controlled. A network event is therefore treated as evidence of suspicious communication, not as proof by itself that an attack has occurred.

- Monitor simulated OT traffic between the virtual PLC, HMI, and other laboratory nodes.
- Identify suspicious or unauthorized control-related communication events.
- Record timestamp, source, destination, protocol, event type, affected asset, and severity.
- Pass normalized network events to the correlation and risk engine.
- Keep all testing inside the isolated virtual laboratory; no real OT or defense-industrial network is involved.

12. Machine-Learning Methodology

The ML experiment follows a baseline-learning approach. Normal telemetry is collected from the simulated process during stable operation. Features are extracted and used to train the anomaly detector. Controlled abnormal telemetry is then presented to the trained model for evaluation.

```
Normal Operation
     ↓
Telemetry Collection
     ↓
Feature Extraction
     ↓
Isolation Forest Training
     ↓
New Telemetry
     ↓
Anomaly Score + Status
```

Random Forest, XGBoost, LSTM autoencoders, and other advanced models are not required for the core prototype. They may be discussed as future work or used as optional comparison models only if the primary pipeline is already stable.

13. Correlation and Risk Methodology

Correlation is based on three practical dimensions: temporal proximity, asset relationship, and event relevance. For example, a suspicious parameter write to PLC-01 followed within a short time window by an anomaly in the physical process controlled by PLC-01 is treated as stronger evidence than either event alone.

The risk score may be represented conceptually as:

$$R = f(SN, SP, CA, SC)$$

where SN represents network severity, SP represents physical anomaly severity, CA represents simulated asset criticality, and SC represents correlation strength. The exact function and weights will be fixed before final evaluation and reported transparently.

The engine should distinguish anomaly detection from security judgment. An ML anomaly means 'unusual process behavior'; a network event means 'suspicious communication evidence'; a correlated high-risk event means that multiple evidence sources jointly justify higher prioritization.

14. Alert Generation and Operator Notification

The alert part is the final visible warning stage of CHEM-SHIELD. It converts the result of the correlation and risk engines into a simple notification that an operator can understand quickly.

An alert is generated when the system has enough evidence to justify attention. For example, if a suspicious network event affecting PLC-01 is followed by an abnormal physical-process state for the same asset within the defined time window, the correlation engine can create a correlated event and the risk engine can assign a higher risk level.

The dashboard can display the alert with the affected asset, time, network evidence, physical anomaly, ML anomaly score, correlation status, risk score, and risk level. The prototype may also show a short non-automated recommendation such as: “Investigate PLC-01 process state and review the related network event.”

The alert mechanism does not automatically shut down the machine, block traffic, change PLC settings, or take control of the system. Automated response is outside the core scope. The purpose of the alert is early warning and operator awareness.

- LOW: informational or low-priority abnormal behavior.
- MEDIUM: suspicious evidence that requires investigation.
- HIGH: strong correlated evidence requiring prompt operator attention.

15. Performance Evaluation

Evaluation will focus on whether the complete pipeline correctly detects and prioritizes controlled abnormal events while avoiding unnecessary alerts during normal operation.

Metric	Purpose
Precision	Measures the proportion of anomaly alerts that correspond to labeled anomalous events.
Recall	Measures the proportion of labeled anomalous events detected by the system.
F1-score	Balances precision and recall.
False-positive rate	Measures unnecessary anomaly alerts during normal operation.
Detection latency	Measures time between the simulated event and the generated alert, where measurable.
Correlation rate	Measures how consistently related network and physical events are combined in the controlled scenarios.

The project should compare at least network-only detection against the combined network-plus-physical approach. Results should be reported for the same controlled scenarios and dataset partitions.

16. Reduced Technology Stack

Component	Primary Choice	Purpose
Programming	Python 3.x	Simulation, data processing, ML, correlation, risk engine
OT Simulation	Python-based virtual PLC/process model	Generate controller, machine, sensor, and telemetry behavior
Network Security	Suricata and/or Zeek	Observe and generate structured network-security evidence
ML	Scikit-learn / Isolation Forest	Physical-process anomaly detection
Data Processing	Pandas / NumPy	Telemetry preparation and feature engineering
Risk/Correlation	Python	Event fusion and risk scoring
Dashboard	Flask + Plotly	Web monitoring interface and visualization
Storage	SQLite	Telemetry/event storage for prototype
Virtualization	VirtualBox or equivalent	Isolation of the laboratory environment

The architecture intentionally allows substitutions. For example, if deploying both Suricata and Zeek proves too resource-intensive, one network-monitoring component can provide the required experimental evidence.

17. Scope, Safety, Ethics, and Dual-Use Controls

The project is strictly defensive and laboratory-based. All abnormal activity is generated inside a controlled virtual environment. No experiment is designed to compromise or disrupt a real system.

- No scanning, probing, or interaction with operational military or commercial networks.
- No connection between the experimental network and live defense/industrial control systems.
- No weaponized malware or persistent offensive tooling.
- No testing against real PLCs, SCADA servers, HMIs, production machinery, or military hardware.

- Synthetic datasets and laboratory configurations are used for evaluation.
- Any simulated attack behavior is limited to what is necessary to generate defensive detection evidence.
- Results should be interpreted as laboratory findings and not as evidence about the security of any real Ethiopian facility.

18. Final Deliverables

10. A functioning isolated OT simulation containing a virtual PLC, machine/process model, and sensors.
11. A network-monitoring component producing structured security events.
12. A baseline telemetry dataset and controlled anomaly dataset.
13. An Isolation Forest-based anomaly detection module.
14. A rule-based event correlation engine.
15. A documented risk-scoring engine.
16. A Flask/Plotly monitoring dashboard.
17. A unified event/data model connecting all components.
18. Experimental results and evaluation tables.
19. Source code, configuration files, sample datasets, screenshots, architecture diagrams, and final thesis documentation.

19. Expected Academic and Practical Significance

The project provides an experimentally reproducible way to study cyber-physical detection without requiring access to real industrial or defense infrastructure. Its academic value comes from evaluating whether evidence from different layers provides a more informative security signal than isolated monitoring.

- Provides a compact OT cybersecurity laboratory suitable for academic demonstration.
- Shows how physical-process behavior can complement network-security evidence.
- Demonstrates transparent evidence correlation rather than treating ML as an unexplained decision-maker.
- Provides measurable performance results rather than only a conceptual architecture.
- Creates a foundation that can later be extended with additional datasets, ML models, protocols, or response mechanisms.

20. Limitations

- The OT environment is simulated and may not reproduce all characteristics of real industrial processes.
- Synthetic telemetry may not capture the full variability of real-world equipment.
- The ML model may be sensitive to the quality and diversity of the baseline dataset.
- Rule-based correlation may not generalize to complex multi-stage incidents.
- Risk thresholds and weights are experimental design choices, not universal standards.
- The results cannot be interpreted as a security assessment of a real Ethiopian defense facility.

21. Conclusion

The revised CHEM-SHIELD proposal preserves the central idea of the original research while reducing the implementation to a realistic, resource-conscious prototype. The project no longer attempts to build a complete industrial cybersecurity ecosystem. Instead, it focuses on one clear research chain:

```
Machine / PLC Simulation
        ↓
Network Monitoring + Physical Telemetry
        ↓
Network Detection + ML Anomaly Detection
        ↓
Correlation of Evidence
        ↓
Risk Prioritization
        ↓
Early-Warning Dashboard
```

Appendix A — Example End-to-End Demonstration

The final demonstration should begin with normal operation and then introduce a controlled simulated abnormal condition. The expected sequence is:

1. The simulated machine operates normally and produces stable telemetry.
2. The network-monitoring component records normal OT traffic.
3. A controlled simulated network event changes a process parameter.
4. The network detector generates a network event.
5. The physical process changes as a result of the simulated condition.
6. The ML engine detects deviation from the learned baseline and produces an anomaly score.
7. The correlation engine associates the network and physical events using time, asset, and event relationships.
8. The risk engine assigns a higher priority to the combined event.
9. The dashboard displays the event, evidence, anomaly score, correlation state, and risk level.
10. The system may display a non-automated response recommendation for operator review.

Appendix B — Recommended Final Project Title

CHEM-SHIELD: An AI-Assisted Cyber-Physical Early-Warning and Threat Detection Framework for Simulated Defense-Industrial Operational Technology

Short title: CHEM-SHIELD

Core focus: AI + Cybersecurity + OT/ICS + Network/Physical Evidence Fusion + Risk-Based Early Warning